Final Project Report: Cooperative Manipulation of Multi-Robot networks using Internal Force Regulation

***Abstract*—The report studies the integration of cooperative object manipulation with implicit control communication. An algorithm is studied for a multi agent system with decentralised control to handle object via implicit coordination. The decentralised leader-follower system is considered where the leading agent has the information of the desired object trajectory following by an impedance control law, while the rest of the agents estimate leaders motion based on error and uses a similar impedance control law. The follower act like a virtual leader for the subsequent agent. The leader guides the motion of object by which followers estimate the motion of object and handle the object accordingly. The torque/force, position and velocity measurements are only the feedback data to the other robot without communicating any other data. Finally the simulation results verify the efficiency of the algorithms.**

***Index Terms*—Cooperative control; infinitesimal rigidity; multi-agent formations; graph rigidity;**

I. Introduction and Motivation

Multi-robot systems have developed a lot over the past decade. Many applications nowadays include robots working autonomously. Generally these robots have a central governing body to command and tell which direction to proceed. This type of body requires a huge amount of computational power especially as the number of entities to control are increased. Focus of current research has been to work on decentralised systems to address these issues. These type of multi-agent systems have a wide range of applications including, formation control, robotic manipulations, rendezvous, etc [1]. These type of systems only communicate with their neighbors and update their own states. The update algorithms are designed to achieve a consensus on the goal.

Robot formation control, every robot tries to maintain a certain formation by communicating only with its neighbors [2]. A neighbor is an agent/robot who is within a communication range. The agents with whom a robot can communicate are denoted by edges in a network. This type of communication is used to arrive at a global consensus for the entire set of agents to form a desired spatial geometric pattern. Here the geometric pattern to be rigid, two main cases are studied, *distance rigidity* and *bearing rigidity* [3]. In rigidity, it is seen for what conditions for a geometric pattern, we can fix the edge length in nodal networks.

Cooperative manipulation is a special type of constrained dynamical system [4–6]. It has majorly been studied in the centralised scenario formats because of its efficiency [7], but is in general less robust and as mentioned earlier has a higher complexity because of its reliance on a central system. Systems are assumed to be attached via rigid joints to the object and hence the complete system is considered to act like a single entity. The control is transmitted without communication in these cases and is mostly done by using force and/or impedance control so the possibility of contact force/torque measured. One of the important property of cooperative manipulation systems by rigid control is that, it can study extensively the effect of internal forces. Agents exert the internal forces on the body and are needed to hold the object. The forces also should not be very low so as to prevent the object from falling and also not very high to damage it by over exertion of force. These forces are exerted on the holding points and they will not have any influence on object motion [8]. Most literature decomposes the forces to motion-inducting and the internal forces, such as [9, 10] decompose the loads on the basis of, can they help in determining the internal force-free expressions, while some are concerned with the cooperative manipulation interaction dynamics.

In past works, forces are decomposed mostly based on agent-agent distances and actual dynamics of the agents are not considered. The report works on integrating rigid cooperative manipulation with rigidity theory, where inter-agent distances and bearings are fixed [8]. The distance rigidity theory and the bearing rigidity theory make study on similar lines if one can determine the unique framework shape from the inter-neighbor distances and inter-neighbor bearings [11]. The report suggests an algorithmic method for decentralized control for object handling by robots and in complete coordination meaning each robot is independent and driven by itself without communicating within robots. The main difference from the conventional algorithms is that it has the application of internal force on the object and also have a specific algorithmic control and motion is purely relied on dynamics.

The report is organized as follows. Problem is formulated in Section II. It provides the necessary preliminaries as well as the model for rigid cooperative manipulation control. Section III discusses the results with the details of distances and bearing rigidity along with the main results. It also gives the simulation results of the test cases. Section IV provides with the conclusion of the topic and Section V gives the direction for future improvements and research.

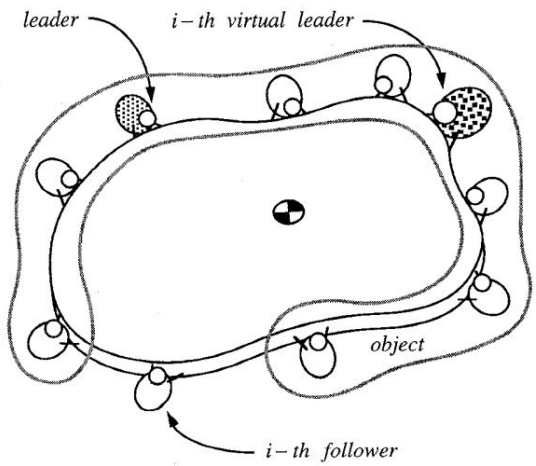

Fig. 1. Robotic agents holding an object [12]

II. Problem Formulation

A. Notation

Natural number are denoted by $\mathbb{N}$, and the n-coordinate space is denoted by $\mathbb{R}^n$. An $n \times n$ identitiy matrix is represented by I_n and the zero vector is represented by 0_n while the $n \times m$ matrix with zero entries is denoted by $0_{n\times m}$. For a matrix $A \in \mathbb{R}^{n\times m}$, A^+ is used to denote its Moore-Penrose inverse [13], null(A), range(A) for the null space and range space. For a discrete set $\mathcal{N}$, $|\mathcal{N}|$ denotes its cardinality. For $a, b \in \mathbb{R}^3$, $S(a)$ is a skew-symmetric matrix defined according to $S(a)b = a \times b$. In addition S^n denotes $n+1$ dimension sphere. $P_r(x) := I_n - \frac{xx^T}{\|x\|^2}$ defines a projection of vector $x \in \mathbb{R}^n$ onto the orthogonal complement of x.

A graph $\mathcal{G}$ is a pair $(\mathcal{N}, \mathcal{E})$ where $\mathcal{N} = \{1, \ldots, n\}$ is a finite non empty node set and $\mathcal{E} \subset \mathcal{N} \times \mathcal{N}$ is an edge set of ordered pairs off nodes called edges. The edge $(i, j) \in \mathcal{E}$ denotes vehicle j can obtain information from i but not necessarily vice-versa. In undirected both graphs edges (i, j) and (j, i) are present. The complete graph of N nodes is denoted by $\mathcal{K}_N$.

The subscripts l is related to leader, f is related to the follower.

Prescribed Performance: Prescribed performance is adopted to characterize the behavior of the error to converge to a predefined set with convergence rate no less than a certain value. If the error is given by $e(t)$, the mathematical expression of performance is given by,

$$-\rho(t) < e(t) < \rho(t), \ \ \forall t \geq 0 \tag{1}$$

where $\rho(t)$ is a smooth and bounded function of time satisfying $\lim_{t\to\infty} \rho(t) > 0$ called the performance function [14]. The exponential performance function is given by,

$$\rho(t) = (\rho_0 - \rho_\infty)e^{-st} + \rho_\infty$$

for appropriate positive constants ρ_0, ρ_∞, s giving performance characteristics to error $e(t)$.

B. Problem Modeling

The multiple robot system will function where the leader has the information of the trajectory to follow and the next subsequent agent follows its trajectory by its own control. This agent also acts like a virtual leader for the next subsequent agent as shown in the figure 2. Hence effectively the system is reduced to solving the control for one follower based on the leader. Hence in the analysis only one leader follower system is considered for simplicity which gives $i \in \{l, f\}$.

For our system in consideration we assume the leader-follower system to have 6 Degrees of Freedom and fully actuated, the desired trajectory $x_{dl}(t)$ is known to the leader and follower estimates it by $x_{df}(t)$ using its own state measurements. As the aim is to have implicit communications, asymptotic tracking is not considered and rather a bound on the error is considered. The assumption taken is that the forces/torques with the object are available for each robot independently.

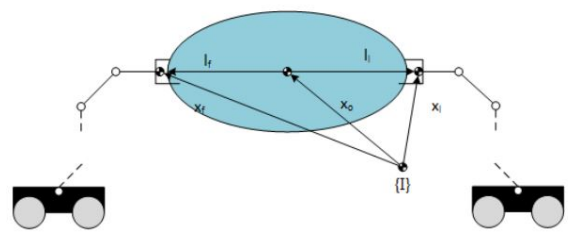

Fig. 2. Robotic agents rigidly holding an object [14]

Kinematics:
The leader and follower's task space (end effector) coordinates with respect to the inertial frame $\{I\}$ are given by, $x_i = [x_{ip}^T, x_{ir}^T]^T, u \in \{l, f\}$, where x_{ip}, x_{ir} correspond to end-effector position and orientation respectively. The object's coordinates are denoted with respect to $\{I\}$ by $x_o = [x_{op}^T, x_{or}^T]^T$. Let also $q_i,\ i \in \{l, f\}$ are the joint space variables. The task space variables as non linear functions of joint variables are expressed by

$$x_i = f_i(q_i),\ i \in \{l, f\}$$

and we also obtain by differentiation,

$$\dot{x}_i = J_i(q_i)\dot{q}_i,\ i \in \{l, f\}$$

where J is the robot Jacobian matrix given by $J_i(q_i) = \frac{\partial f_i(q_i)}{\partial q_i}$.
For a N robotic agent system $\mathcal{N} = \{1, \ldots, N\}$ rigidly holding an object, $q_i, \dot{q}_i \in \mathbb{R}^{n_i}$, $n_i \in \mathbb{N}, \forall i \in \mathcal{N}$ are the generalised joint-space variables and their time derivatives. $q = [q_1^T, \ldots, q_N^T]^T$. The inertial position and orientation of the ith end-effector is given by $p_i \in \mathbb{R}^3$ and $\eta_i \in \mathbb{T}$, $\mathbb{T}$ being the orientation space. Velocity of the end-effector is given by, $v_i := [\dot{p}_i^T, \omega_i^T]^T$, ω being the angular velocity in $\mathbb{R}^3$. $v_i - J_i(q_i)\dot{q}_i$. Since the system contact are considered rigid, the following relations hold:

$$x_{ip} = x_{op} + l_i,\ x_{ir} = x_{or} + a_i,\ i \in \{l, f\} \tag{2}$$

where the vectors $l_i = [l_{ix}, l_{iy}, l_{iz}]^T$ and $a_i = [a_{ix}, a_{iy}, a_{iz}]^T$ represent the relation of object and the end-effector's frame.

As we already know the geometric parameters, each agent can compute its objects coordinates and we get velocity relation by differentiating (2)

$$x_i = J_{oi}\dot{x}_0,\ i \in \{l, f\} \tag{3a}$$

$$\ddot{x}_i = \dot{J}_{oi}\dot{x}_o + J_{oi}\ddot{x}_o,\ i \in \{l, f\} \tag{3b}$$

Here J_{oi} is the Jacobian from the end-effector to the objects center of mass. Since the object and robot are rigidly connected, we can say Jacobian always has a full rank and hence has a well defined inverse J_{oi}^{-1}. Differentiation (3a) we get the equation (3b).

Dynamics:
The joint and task space dynamics of single agent is:

$$B_i(q_i)\ddot{q}_i + N_i(q_i, \dot{q}_i)\dot{q}_i + G_{q_i}(q_i) = \tau_i + J_i(q_i)^T F_i, \tag{4a}$$

$$M_i(q_i)\ddot{x}_i + C_i(q_i, \dot{q}_i)\dot{x}_i + G_i(q_i) = U_i + F_i,\ i \in \{l, f\} \tag{4b}$$

where B_i is the positive definite inertia, N_i is the Coriolis force and G_{q_i} is the gravity term, $M_i := (J_i B_i^{-1} J_i^T)^{-1}$, $C_i\dot{x}_i := M_i(J_i B_i^{-1} N_i - \dot{J}_i)q_i$, and $g_i := M_i J_i B_i^{-1} g_{q_i}$. The vector $F_i,\ i \in \{l, f\}$ represents the interaction force/torque exerted at the end-effector by object and $U_i,\ i \in \{l, f\}$ denotes the input task space wrench [14]. The relation for torque joints τ_i and the task space wrench is as follows

$$\tau_i = \bar{J}_i^T U_i + (1 - J_i^T \bar{J}_i^T)\tau_{in},\ i \in \{l, f\}$$

where $\bar{J}_i$ is considered the generalised inverse consistent with the equations of motion for the manipulator. The vector τ_{in} is not responsible to the end-effectors wrench and is analysed separately for other goals such as collision avoidance. Thus we can express the dynamic models with respect to the objects coordinates as follows:

$$M_{oi}(q_i)\ddot{x}_o + C_{oi}(\dot{q}_i, q_i)\dot{x}_o + G_{oi}(q_i) = K_{oi}^T U_i + J_{oi}^T F_i \tag{5}$$

$\forall i \in \{l, f\}$ where $M_{oi}(q_i) = J_{oi}^T M_i(q_i) J_{oi}$, $G_{oi}(q_i) = J_{oi}^T G_i(q_i)$, $C_{oi}(\dot{q}_i, q_i) = J_{oi}^T(C_i(\dot{q}_i, q_i)J_{oi} + M_i(q_i)\dot{J}_{oi})$
Similarly the dynamic equation of object is:

$$M_o(x_o)\ddot{x}_o + C_o(\dot{x}_o, x_o)\dot{x}_o + G_o(x_o) = F_o \tag{6}$$

It is to be noted here that if no external forces are exerted, then the total force $F_o = -GF$ where $G = [J_{ol}^T, J_{of}^T]$ denotes the grasp matrix for the full configuration and $F = [F_l^T, F_f^T]$.

We further divide the forces exerted by the end-effectors into the internal and motion-induced forces: $F_i = F_i^m + F_i^{int}$. The internal forces on the system belong to the null space of the grasp matrix G and do not contribute towards the acceleration of the object.

C. Distance and Bearing Rigidity

We define a function γ that encodes the distance and bearing constraints, γ_d and γ_b of a rigid system as follows:

$$\gamma_d(x_{ip}, x_{jp}) = \frac{1}{2}\left\|x_{ip} - x_{jp}\right\|^2, \tag{7a}$$

$$\gamma_b(x_i, x_j) = x_{ir}^T \frac{x_{ip} - x_{jp}}{\|x_{ip} - x_{jp}\|}, \tag{7b}$$

where $\|x_{ip} - x_{jp}\|$ is the distance between agents and $x_{ir}^T \frac{x_{ip} - x_{jp}}{\|x_{ip} - x_{jp}\|}$ is the local bearing vector expressed in agent i's frame. The rigidity function γ is formed by stacking the distance and bearing functions:

$$\gamma = \begin{bmatrix} \gamma_d(x_p) \\ \gamma_b(x) \end{bmatrix} \tag{8}$$

The internal motions that preserve the rigidity of the system are known as *infinitesimal motions*. These are motions that do not alter γ. Hence, we define the *rigidity matrix* as $R = \nabla\gamma$, so that the desired infinitesimal motions belong to the null space of this matrix. We have,

$$R = \begin{bmatrix} \frac{\partial\gamma_d}{\partial x_p} & \frac{\partial\gamma_d}{\partial x_r} \\ \frac{\partial\gamma_b}{\partial x_p} & \frac{\partial\gamma_b}{\partial x_r} \end{bmatrix} \tag{9}$$

As shown in [15], the system conserves rigidity if,

$$\text{rank}(R) = \dim(R) - \dim(\text{null}(R)) = 6N - 6 \tag{10}$$

Also, to avoid internal forces that violate the rigidity constraints, the null space of G should be equal to the range of R, which gives us feasible control.

III. Control Methodology

A. Impedance Control Scheme

The inertial and geometric parameters of both mobile manipulators and object are known hence a feedback linearization scheme is implemented as in [14] for each robot. In this respect teh following inputs are considered:

$$U_i = -F_i + J_{oi}^{-T}(M_{oi}V_i + C_{oi}\dot{x}_o + G_{oi}),\ i \in \{l, f\} \tag{11}$$

To remove the non linearities of (4) $V_i,\ i \in \{l, f\}$ is chosen as follows:

$$V_i = \ddot{x}_{cmd,i} + M_o^{-1} J_{oi}^T(F_i - F_{di}),\ i \in \{l, f\} \tag{12}$$

Thus the desired impedance behavior is achieved, $\ddot{x}_o = \ddot{x}_{cmd,i} + M_o^{-1} J_{oi}^T(F_i - F_{di})$ where $F_{di},\ i \in \{l, f\}$ denotes the desired robot/object interaction wrench:

$$F_{di} = F_{int,i} - J_{oi}^{-T} c_i(C_o\dot{x}_o + G_o + M_o\ddot{x}_{cmd,i}) \tag{13}$$

This selection is done as it cancels the object's non linearities and ensures adequate internal forces and achieves control objectives. Moreover, the load distribution coefficients, $c_i, i \in \{l, f\}$, and design constraints $c_l + c_f = 1,\ c_i, c_f > 0$ are assigned according to payload capabilities according to the amount a robot can pick in heterogeneous systems. For actual objects desired trajectory profile to be uncorporated by the leader and estimate by the follower given by $x_{dl}(t)$ and $x_{df}(t)$ respectively the leader and follower tracking error dynamics are given by [14],

$$\Delta\ddot{x}_i + D_i\Delta\dot{x}_i + K_i\Delta x_i = M_o^{-1} J_{oi}^T(F_i - F_{di}) \tag{14}$$

here $_i = x_o - x_{di}\ i \in \{l, f\}$ For convenience and simplicity we choose $D_l = D_f = D$ and $K_l = K_f = K$ as the object dynamics and hence we get,

$$\Delta\ddot{\bar{x}} + D_i\Delta\dot{\bar{x}} + K_i\Delta\bar{x} = 0 \tag{15}$$

We define $\Delta\bar{x} = x_o - \frac{(c_l+1)x_{dl}+(c_f+1)x_{df}}{3}$. Hence because of the control gain matrices being positive definite, the system becomes asymptotically stable. Hence $\Delta\bar{x}$ and $\Delta\dot{\bar{x}}$ converge to zero exponentially. Hence,

$$x_o(t) = \frac{c_l + 1x_{dl}(t) + (c_f + 1)x_{df}(t)}{3} + \Delta\bar{x}(\Delta\bar{x}(0), \Delta\dot{\bar{x}}(0); t) \tag{16}$$

We see, $\Delta\bar{x}(\Delta\bar{x}(0), \Delta\dot{\bar{x}}(0); t) \to 0$ exponentially.

Estimation Law

The follower agent is unaware of the actual trajectory that the leader plans to follow. There is not communication between the agents directly, but the follower can estimate the error value by measuring the term $3\frac{x_o(t)-x_{df}(t)}{c_l+1}$. Since acceleration measurements are not available by the agents, the asymptotic stability is not considered and instead a bounded error is defined. As defined earlier in (1) each element of error $e(t)$ has to follow the inequality;

$$-\rho_j(t) < e_j(t) < \rho_j(t),\ \forall t \geq 0 \tag{17}$$

The candidate performance function would be given by,

$$\rho(t) = (\rho_0 - \rho_\infty)e^{-st} + \rho_\infty$$

the constant s dictates the exponential convergence rate and $\rho_{j,\infty}$ denotes the final boundedness. The $\rho_{j,0}$ is chosen to satisfy $\rho_{j,0} > |e_j(0)|$. The estimation law is designed [14]:

$$\dot{x}_{dfj} = k_j ln\left(\frac{1 + \frac{e_j(t)}{\rho_j(t)}}{1 - \frac{e_j(t)}{\rho_j(t)}}\right) \quad k_j > 0$$

B. Stability Analysis:

For a given error $e(t) = [e_1(t), \ldots,]^T = 3\frac{x_o(t)-x_{df}(t)}{c_l+1}$ considering a smooth $x_{dl}(t)$ with bounded derivatives as well as some appropriately selected performance functions $\rho_j(t)$ satisfying $|e_j(0)| < \rho_j(0)$, the estimation law guarantees that error is bounded by $\pm\rho_j(t)$

Proof: From (16)

$$|e_j(t)| = \left| x_{dlj}(t) - x_{dfj}(t) + \frac{3\Delta\bar{x}_j(\Delta\bar{x}_j(0), \Delta\dot{\bar{x}}_j(0); t)}{c_l + 1} \right| < \rho_j(t)$$

This implies

$$|x_{dlj}(t) - x_{dfj}(t)| < \rho_j(t) + \frac{3|\Delta\bar{x}_j(\Delta\bar{x}_j(0), \Delta\dot{\bar{x}}_j(0); t)|}{c_l + 1}$$

And as we already know that $\Delta\bar{x}(\Delta\bar{x}(0), \Delta\dot{\bar{x}}(0); t) \to 0$ exponentially, the value is untimately bounded to ρ_∞. The tracking error is bounded therefore in the range $\frac{c_f+1}{3}$

IV. Simulation Results

A 1-D scenario is considered for the two robots and object handling by leader-follower scheme is used. A sinusoidal path trajectory is assigned to the leader and follower moves by estimating it without communicating to leader using the proposed algorithm. These are the parameters used in algorithm:

Quantity	Value
$c_l = c_f$	0.5
D	2
K	1
k_1	0.5
a	2
b	1

The value for $\rho_1(t) = 0.49e^{-t} + 0.01$ and $F_{int,i},\ i \in \{l, f\}$ were set to zero.

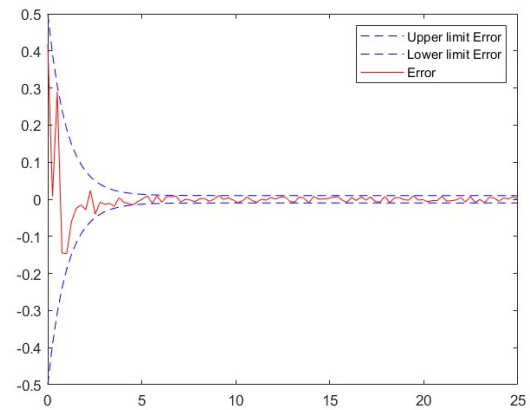

Fig. 3. Error Estimation along with $\rho(t)$ bound

As we observe in the simulation result, the estimation error and the proposed limits, tell us the control algorithm keeps the values of error bounded within the range specified by $\rho(t)$. We also see that the value of $\rho(t)$ tends exponentially to a value close to zero. It is upon the users discretion on how accurate the final ρ_∞ is needed although the resulting of measurement instruments are needed to be kept in mind.

V. Scope of Work

Based on the concepts in report, there are a number of applications for which this type of system would be quite useful. Some of them are listed as follows:

(i) *Space Robotics:*
Considering the current scenario, with increasing structures in orbit around earth, the size is has limitation. We are limited by the size of object we can put into orbit because of our launch rocket size being limited. Also space debris are another emerging issue because of increased number of launches. A set of satellites with robotic arms, could be aimed to target these debris to bring them back for re-usability. Also, the satellites could be multi functional and helpful in space assemblies. In these scenarios, the multi-agent approach stated to hold the concerned body could be very useful as they will reduce the communication load and at the same time increase autonomy to the function.

(ii) *Structure Transfer:*
With the increase in size of structures made, to transport them from one place to another requires an even larger base or stable support structure. Applications of these

type are transfer of huge pipes, etc. For these type of cases, multi-agent approach to transfer these structures using a combination of robots with holding arms, could reduce the size of stable structure to be made and give the same level of accuracy to move the structures from one point to another.

(iii) *Aerial Robotics:*
Aerial Transport systems for payloads towed by cables is generally seen in an emergency response systems as well as industrial and military applications [16]. Most of these applications include transporting the object/payload to regions which might be inaccessible by other means. Currently, these applications are achieved by towing the object by a single aerial vehicle, which requires huge amount of power and thus a limitation on the size of payload to be transferred. But with the current approach multiple agents could work on the same and have much larger payloads by the multi-agent approach.

(iv) *Medical Applications:*
Applications in micro surgeries where the devices used are very small in size, have the risk of puncturing blood vessels during surgeries. This type of sensitive control algorithm can act like a feedback control to the surgeon while holding any sensitive blood vessels.

(v) *Automotive industry:*
One of the interesting applications to be considered could be in the automotive sector. Here the front tires act like the leader and rear tires optimise their suspension according to the response the car (rigid body) in contact with both the tires.

VI. Conclusion

In this report a cooperative manipulation strategy is provided for a leader-follower robot configuration. The agents are able to cooperatively maneuver an object without any explicit communication between them. The only interaction between the agents is implicit, i.e., through the internal forces sensed by each agent. An impedance control law is implemented by the leader to move the object to a desired pose. The follower then uses an estimator to update the desired trajectory of the object, which is originally unknown to the follower. The follower is also able to control the object through an impedance law. As shown in the results, the final error are bounded by an exponentially decreasing functions. The bounds are dependent on the defined system parameters. We also relate the constraints of the system to the bearing and rigidity constraints, which provide us with a span of feasible control efforts, such that the internal forces in the system are minimized.

VII. Ideas of Further Improvements

The report gives a comprehensive way on how the distance and bearing rigidity matrix allow coordinated motions of multi-agent with object systems. Future efforts could be more focused towards minimization of the arising internal forces. This would give an estimate of the amount of forces that the body needs to exert to not only understand the changes in motion for the stabilisation algorithm to work, but also the hold the object.

Another interesting direction of research could be to understand the influence of external disturbances in the system. In the current scenario, only forces because of internal robots manipulation are considered. If external forces are introduced then how the system is modified and will the convergence still occur under any conditions is another direction to explore. This scenario would also include having the control law to incorporate the possibility of obstructions. In current analysis since only two agents were considered, the possibility of having them interfere was not considered, but in multi-robot system, the interference could be an issue needed to be addressed.

The system only considered the object manipulations and not the robotic manipulations. Most of the robots with this application are aimed to carry an object from one point to another. Hence the next work could be more focused towards, how this object manipulation could work with robots having their own dynamic constraints. It could also include optimisation of velocity of the object to the specific applications.

Another interesting approach would be to study if the follower robots can learn the trajectory via ML algorithms. In such a case if the leader is implementing say a dagger algorithm, then if the implicit learning is able to teach the other following robots, the computational and communicational load to teach other agents would be reduced significantly.

The basic assumption for the system taken into analysis in our case was that the object was rigid. A comprehensive study towards the direction of having this kind of control algorithm for a non rigid object could make the algorithms applications more wider and therefore better suited.

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